

AI ENTERPRISE SUMMIT · CHARLOTTE · OCTOBER 8, 2026

# Previously Impossible

Owen Parker

Chief Technology Officer

Kurtis Cochran

Chief Financial Officer



# More people live with kidney disease than with cancer.

37M

US adults living with chronic kidney disease, about 14%

18.6M

Americans living with a history of cancer

No cure

for chronic, scarring kidney disease. A transplant treats kidney failure. It is not a cure.

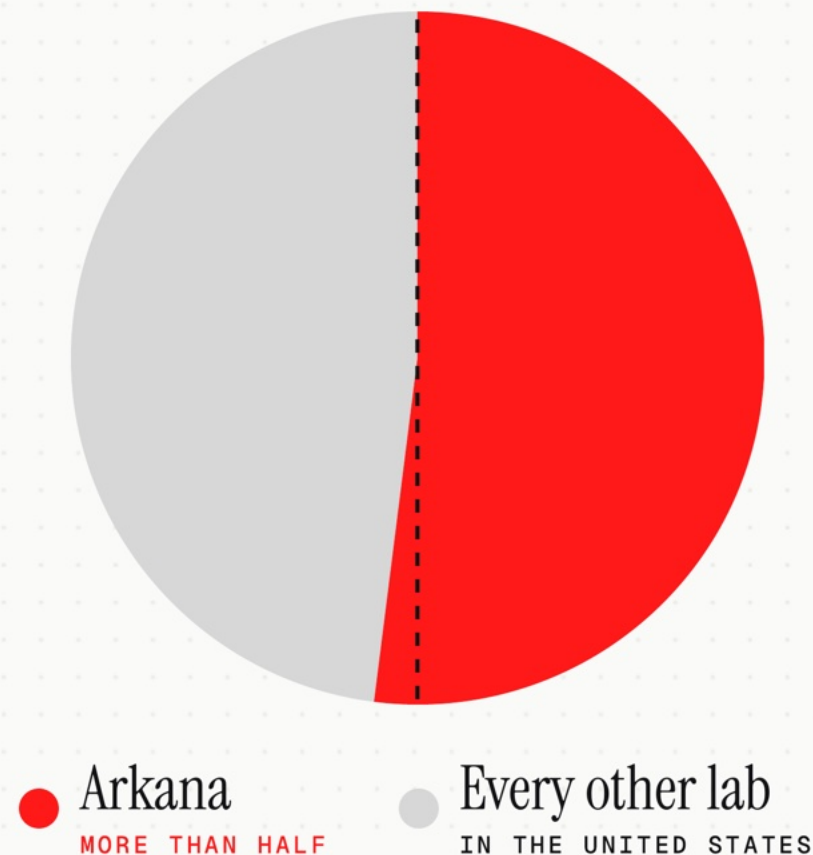
Sources: CDC, Chronic Kidney Disease in the United States, 2026. Wagle et al., CA: A Cancer Journal for Clinicians, 2025. NIDDK, Kidney Transplant.

OUR MISSION

Improve patient care by advancing understanding of disease and providing world-class diagnostics.

## OUR SCALE

A 250-person lab reads more than half of the kidney biopsies in the United States.



Estimated share of all US kidney biopsies, native and transplant. Arkana internal estimate.

# Every transportation evolution forced a choice.

## Horse

GAINED

Goes anywhere

GAVE UP

Distance and load

## Railroad

GAINED

Far and heavy

GAVE UP

Only where the track is

## Car and truck

GAINED

Door to door

GAVE UP

The efficiency of rail

1908 First Model T    1918 US farm horses peak

*Each transition took decades. The holdouts  
had a real argument.*



THREE LAYERS OF VALUE

# Efficiency AI is table stakes.

LAYER 1 · EFFICIENCY

Summarize my meeting.  
Draft this document. Build  
me this slide.

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Table stakes. Everyone has it.

LAYER 2 · REBUILD

Tear up the track. Rebuild  
a core process with AI at  
the center.

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At Arkana: billing code automation

LAYER 3 · GREENFIELD

Build the routes nobody  
thought were buildable.

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At Arkana: Connect, and our core lab  
system

## THREE CLAIMS

# 1 Efficiency AI is table stakes.

The greater value is in rebuilding core processes, and in greenfield.

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# 2 Small and regulated is an advantage, not a reason to wait.

Building internally turned out to be faster, more compliant, and more capable.

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# 3 Greenfield compounds.

Systems you own capture data no one else has. That data feeds your models, and the moat widens.

TWO YEARS

# What a 250-person lab built in-house

## REVENUE CYCLE

### Billing code automation

Fixes stain billing codes in transit

## INTAKE

### Digitized accessioning

Morning paperwork scanned, parsed, and extracted by AI

## PHYSICIANS

### Templatized reporting

Dictated audio to consistent, templated reports

## LOGISTICS

### Next-day intake prediction

Forecasts tomorrow's package intake

## BUILT BY OPERATIONS

### Contract management

Replaced external software at near-zero marginal cost

## OUR OWN DATA

### Disease-identification models

Early: tuned on our own kidney pathology data

## PHYSICIAN APPLICATION

### Arkana Connect

Brought in-house, stabilized, and expanded

**All of it in-house.**

Under two years. Engineering and data team: fewer than six for most of it.

# Every lab stain needs the right billing code. Staff used to fix each one by hand.

01 A case is logged in the lab system (LIS) and stains are ordered.

| *LIS converts the order into an HL7 charge message and sends it to the billing system.*

02 Every stain type bills under its own CPT codes. The first stain gets one code and each additional stain gets a different code, and most cases have several stains of each type.

03 The billing system can't split stain charges into the correct first/additional CPT codes, so it drops them into a work queue for the RCM team.

04 RCM staff manually open each case to change the CPT codes by hand, taking 5 min per case on 150 cases per day.

05 The corrected claim is sent for billing.

## WHY IT WAS MANUAL

The uniqueness of our billing needs created too little an economic incentive for billing providers to build it into their system, leaving us without a scalable solution.

## KEY TERMS

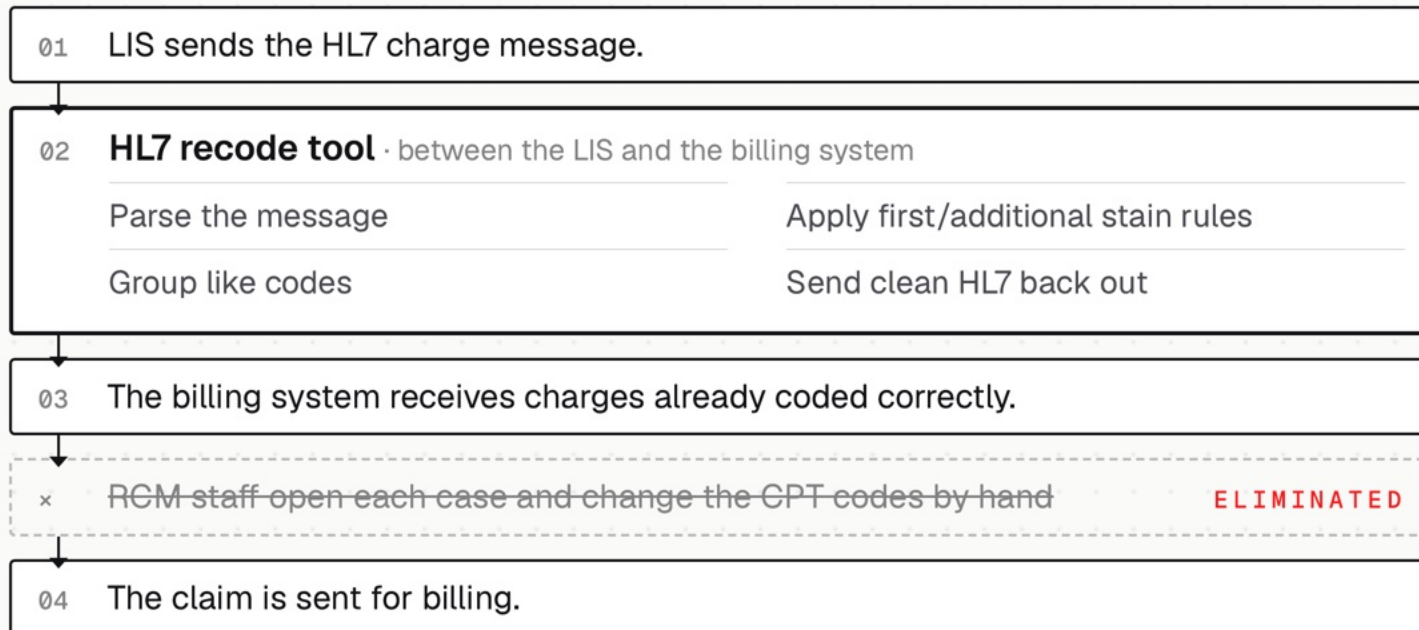
**LIS** · the lab system

**HL7** · the message format systems use to send charges

**CPT** · the code that tells a payer what was done

**RCM** · the revenue cycle team

# The HL7 recode tool fixes codes in transit. The hand-correction step is gone.



## ONLY HUMAN TOUCH

A person steps in only when a new stain order code is added to the LIS.

## STATUS

Tested against de-identified cases; not yet in production.

## What the tool's own log shows.

26.6

HOURS SAVED

531 cases in, this returned 26+ hours of manual coding time with zero exceptions and zero QC failures, and building it surfaced a duplicate-billing bug that was costing us before the tool existed.

531

cases where the tool changed a charge; 80 in the most recent week

1,076

cases sent to billing across 7 batches

0

cases that needed a person to review

0

errors found in 210 spot-checked cases

KURTIS COCHRAN · CHIEF FINANCIAL OFFICER

# What is the next layer of your team's work?

GREENFIELD

# Previously impossible.

Not technically impossible. Unapproachable from the chair you sit in.

ARKANA CONNECT

## Our reports

How nephrologists receive them.

VENDOR BUILD

## About two years

Most of the way there.  
Never finished.

BROUGHT IN-HOUSE

## Finished

Stabilized and expanded  
by our own team.

PHYSICIAN ENGAGEMENT

4 to 5x

Since it came in-house.

## OUR MOST ACTIVE INITIATIVE

# For almost twenty years, we bent the business around a box in a closet.

### WHERE THE WORK ACTUALLY LIVED

- Case counts in Teams messages
- A shared inbox for stain orders
- Spreadsheets for case finalization
- Paper
- Clipboards

### ONE PIECE AT A TIME

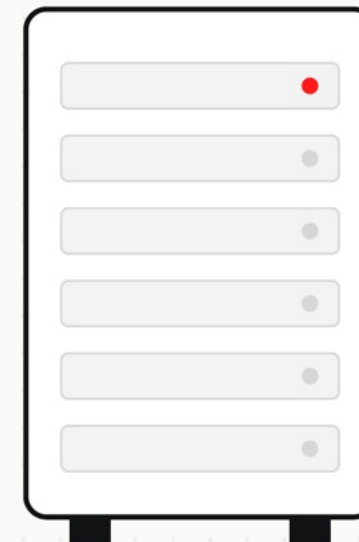
#### Report builder

Dictated audio to consistent, templated reports.

#### Accessioning

The morning's paperwork, scanned, parsed, and extracted by AI agents.

*Each piece does something the box could not.*



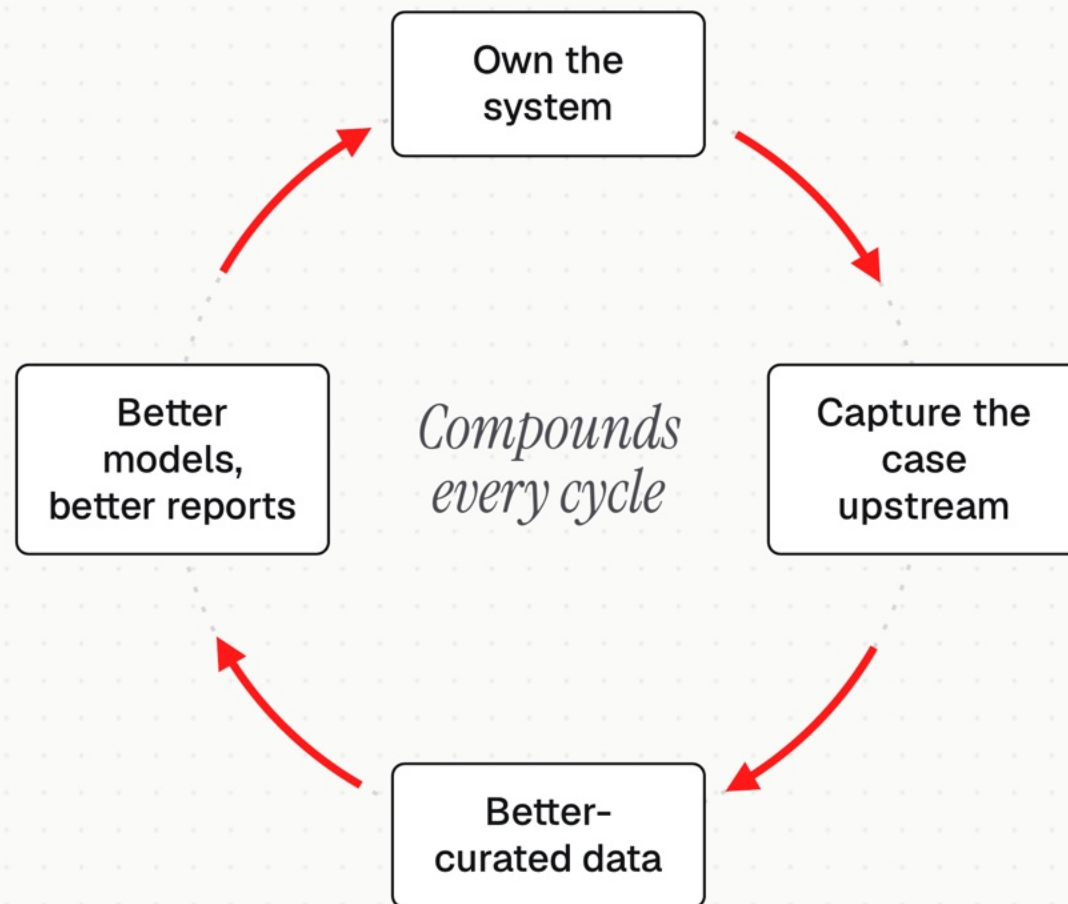
THE THREAD UNDER ALL OF IT

# Capture data upstream.

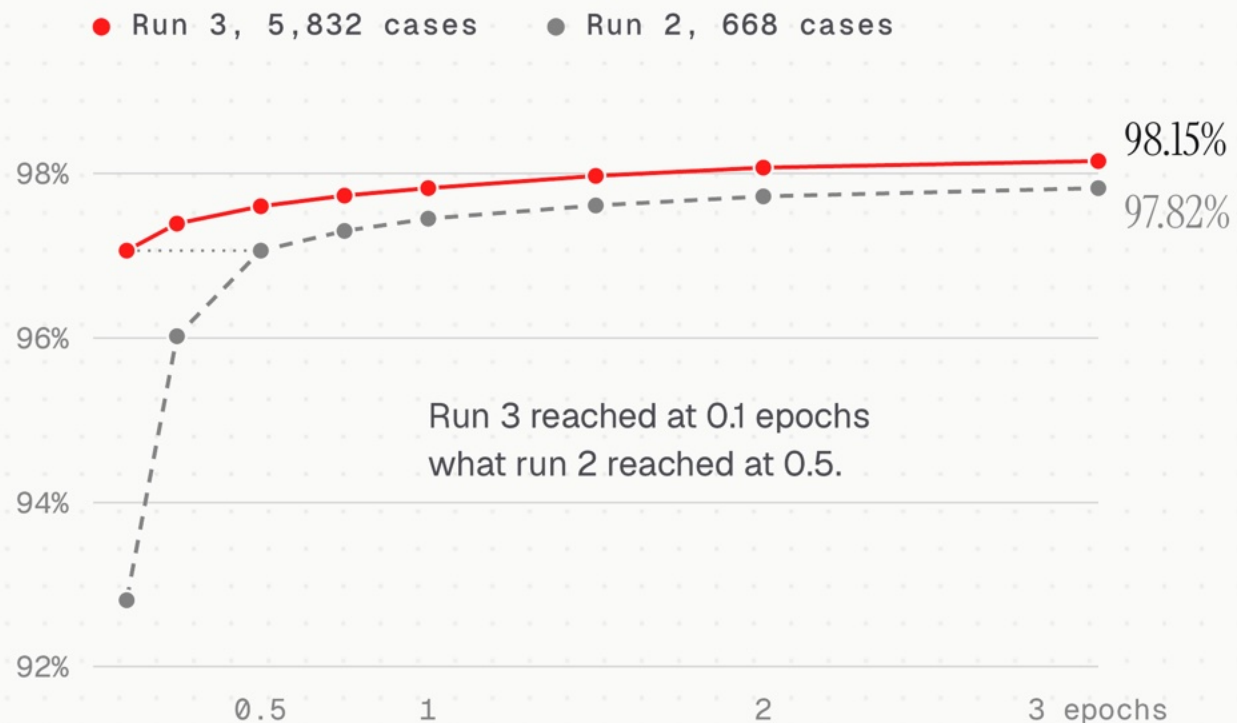
The same system produces a better diagnostic report for the patient and a better-curated dataset for our models.

THE OLD BOX

WHAT WE ARE BUILDING



# Nine times the data. Every code covered.



## CASES

668 → 5,832

About nine times the data.

## DIAGNOSTIC CODES COVERED

178 of 178

Up from 115, or 65 percent.

## NEXT-TOKEN ACCURACY

98.15%

Up from 97.82%. Not diagnostic accuracy.

## THE LONG TAIL

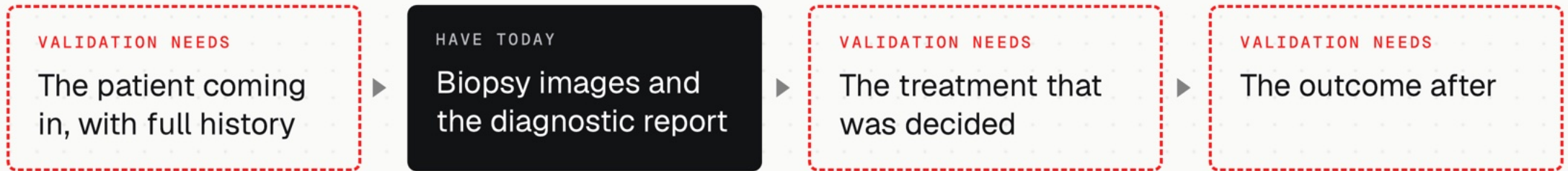
62 codes

Still have five or fewer examples.

Two fine-tuning runs on our own cases, held-out evaluation. Not validated for clinical use.

UNFINISHED

# Next-token accuracy is not diagnostic accuracy.



Millions of slides. Physical tissue we can always go back to, scan again, and keep the flywheel turning.

## ENABLEMENT

# None of this came from tools.

It came from curiosity, and a seven-week cohort.

### WHO

## Twelve people

One cohort, drawn from teams throughout the company.

### HOW LONG

## Seven weeks

Two hours a week.

### WHAT

## One book

*Vibe Coding*, by Gene Kim and Steve Yegge.

### HOW

## Real problems

Experimentation with real direction.

Curriculum built in-house by Sadie, one of our leaders in AI adoption.

# AI compresses the middle of work. Value migrates to the edges.

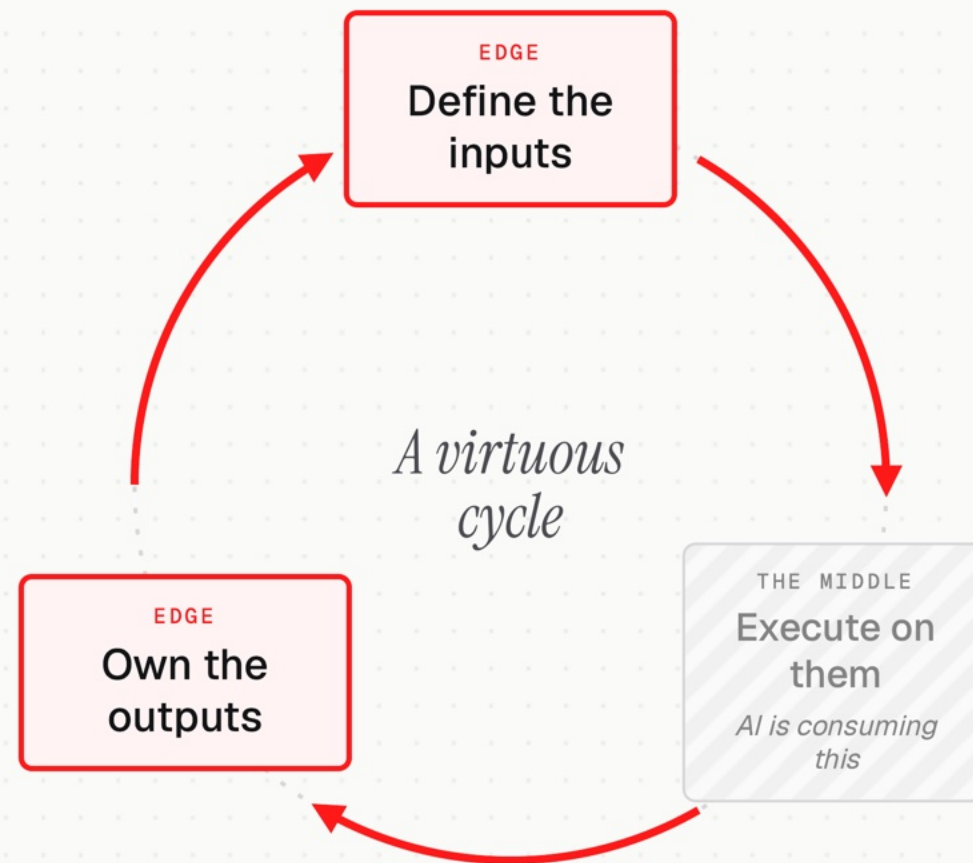
FOR A PATHOLOGIST

**EDGE** Deciding what to look for in the biopsy

**MIDDLE** Pages of paperwork, read for the clinical history

**EDGE** Assigning the diagnosis

*Not just for pathologists. The value is migrating to the edges in every role at Arkana.*



TAKE HOME

# Two things for Monday.

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IF YOU ARE AN ENGINEER

Find a small company with a genuinely hard, specific problem.

The kind off-the-shelf software has never been able to solve. That is where a lot of the real work is, and more of the upside.

IF YOU RUN ONE

Stop traversing the track somebody else laid.

The vehicle is no longer the constraint. Neither is the track. Go build the things you thought were previously impossible.

## THE ASK

# What we need help with.

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### TALENT

We're hiring across the business.

Engineering, operations, and the lab. If a hard, specific problem is what you want to work on, come build it with us.

### DATA

A longitudinal dataset is the foundation.

These problems unfold over years, not in a single biopsy. Building that record over time is how they get solved, and we can't build it alone.

### CONNECTIONS

Advisors and collaborators.

People and institutions with the expertise and resources to work on this with us. Introductions are welcome.

# Being unique isn't rare. Building for it is.

And if you want to, you're welcome to come build with us.



THANK YOU · OWEN PARKER · KURTIS COCHRAN

# Sources

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10. Fine-tuning results: Arkana internal training comparison, run 2 versus run 3, April 2026.
11. Kim G, Yegge S. Vibe Coding. IT Revolution Press, 2025.

